DAURES GREEN HYDROGEN VILLAGE

University of Stuttgart
IER Institute of Energy Economics and Rational Energy Use

With funding from the:
Federal Ministry of Research, Technology and Space

# Transporting Feedstock for eFuel Production in Namibia

## BACKGROUND

Supplying an eFuel plant requires two essential feedstocks: a concentrated carbon source (typically $CO_2$) and hydrogen ($H_2$). In geographically large, infrastructure-sparse regions, the cost and practicality of moving these molecules can become a defining constraint on whether a project is scalable beyond the pilot stage. This fact sheet summarizes a techno-economic comparison of transporting hydrogen versus carbon dioxide over a 780 km corridor route Daures – Otjikoto Power Plant in Namibia (1560 km round trip) for a sustainable aviation fuel (SAF) production configuration.

The central conclusion of the assessment is that across the evaluated throughputs and sensitivity cases, transporting $CO_2$ is consistently cheaper than transporting $H_2$, primarily because hydrogen logistics are dominated by high energy penalties (especially for liquefaction) and low payload densities per vehicle. As a result, the economically preferred configuration is to co-locate eFuels facility near the green hydrogen plant and transport only $CO_2$.

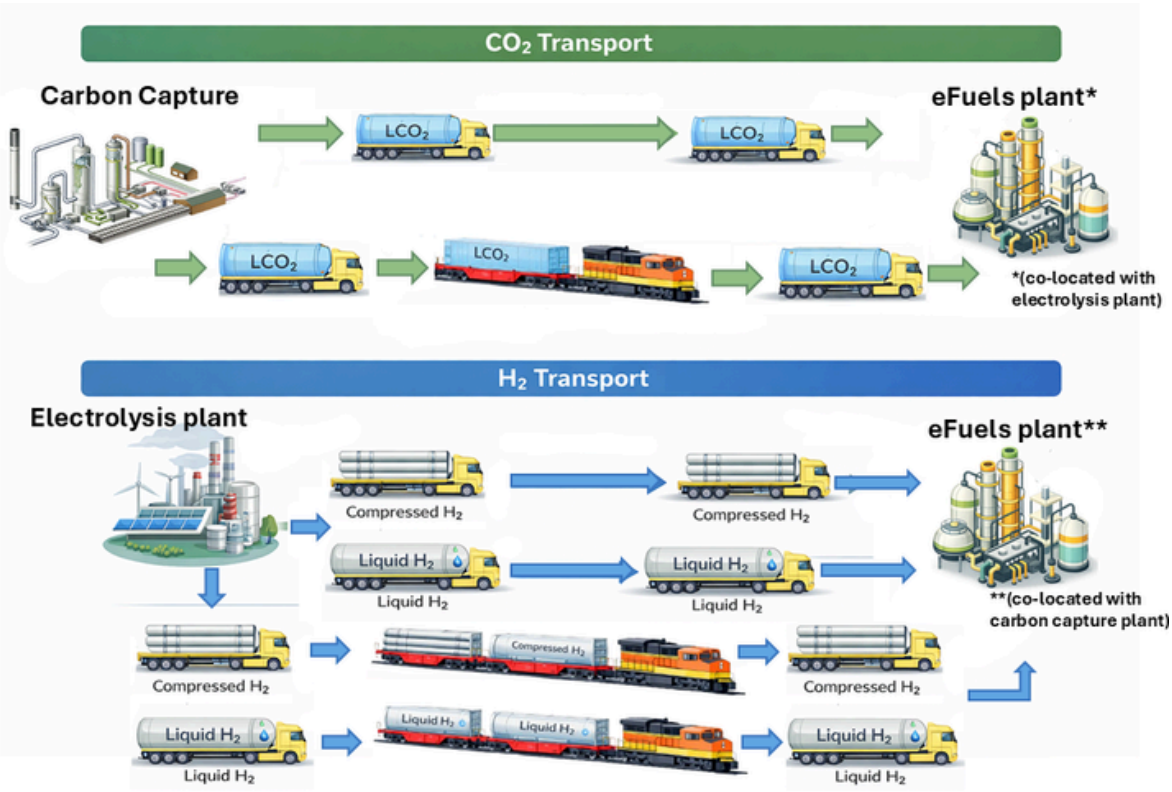


Figure 1: Considered scenarios in the transport feasibility studies

At first glance, transporting $CO_2$ may seem disadvantageous because more $CO_2$ mass is required than $H_2$ mass to synthesize hydrocarbons. The analysis uses a stoichiometric ratio of 7,2 kg $CO_2$ per kg $H_2$, which means that, on an energy basis, the $CO_2$ mass needed is approximately 7,2 times the hydrogen mass required to produce 1 MJ of SAF. However, mass alone is not the cost driver. The decisive factors are (i) the energy required to condition each molecule into a transportable form, and (ii) how much of that conditioned product fits into a practical transport unit.

In the assumptions used here, liquefaction energy for hydrogen is 9 – 10 kWh per kg $H_2$ [1], whereas $CO_2$ liquefaction is 0,09–0,13 kWh per kg $CO_2$ [2]. Payload density amplifies this difference. Typical trailer payloads applied in the report are around 0,8 t $H_2$ for compressed gas tube trailers, 3,6–4 t $H_2$ for liquid hydrogen trailers, and around 22 t $CO_2$ for a cryogenic ISO tank container. This creates a structural logistics advantage for $CO_2$: fewer vehicle movements are required to deliver the same "chemical function" to the plant, even though more $CO_2$ mass is needed.

## SCENARIO ANALYSIS

The comparison includes amortized capital costs for transport equipment, operating costs (diesel, labour, and power for compression/liquefaction and storage), and fleet sizing based on turnaround cycles for the 780 km.

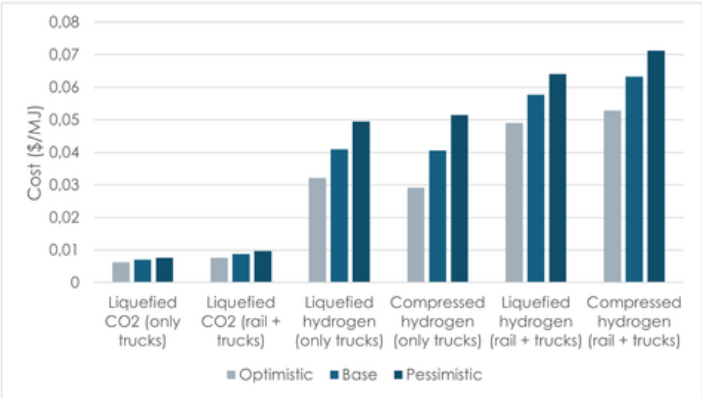


Figure 2: Cost comparison of Liquid CO2 with Liquid H2 and Compressed H2 to produce 1MJ SAF, in different scenarios and different means of transport.

The assessment shows large gaps between $CO_2$ and $H_2$ transport costs in the Namibian configuration, when conditioning and storage costs are considered.

- $CO_2$ trucking: 0,06 – 0,07 $/kg
- $CO_2$ trucking and rail transporting: 0,08 – 0,09 $/kg
- Compressed $H_2$ trucking: 2,1 – 2,8 $/kg
- Liquid $H_2$ trucking: 2,3 – 2,67 $/kg
- Compressed $H_2$ trucking and rail transporting: 3,7 – 3,9 $/kg
- Liquid $H_2$ trucking and rail transporting: 3,4–3,5 $/kg

Rail is evaluated where applicable, with special attention to the "double-handling" costs that occur when trucking must still be used to bridge to and from rail terminals.

The evaluated throughput range is aligned to the staged project Daures logic (Phase 1.5 through Phase II), between 18.000 – 64.000 t/y $H_2$ and the corresponding stoichiometric CO2 demand of 129.600 – 460.800 t/y $CO_2$.

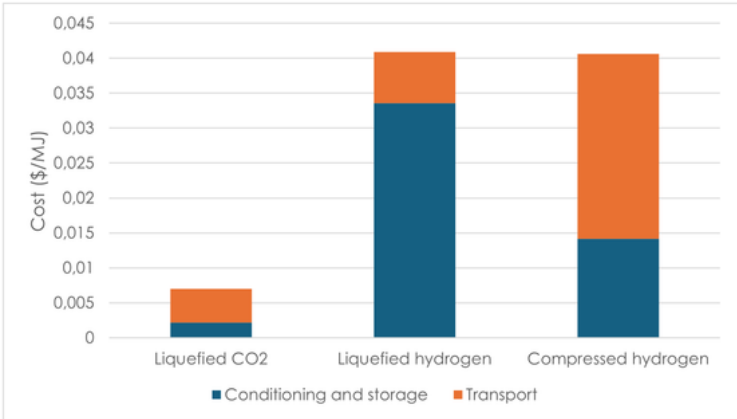


Figure 3: Cost comparison of cost structure of LCO2 with LH2 and CH2 to produce 1MJ SAF, in base case scenario and trucks only.

These values align closely with the broader literature and are not marginal effects.

Even though $CO_2$ mass flow per MJ SAF is larger, transporting $CO_2$ is still 3,8 to 7 times cheaper to move than hydrogen for the SAF production in the context considered, because hydrogen liquefaction imposes an energy penalty consuming between 35 – 45% of the lower heating value of hydrogen [3], and hydrogen high transport equipment and logistics scale unfavorably.

## SCALABILITY AND OPERATIONAL FEASIBILITY

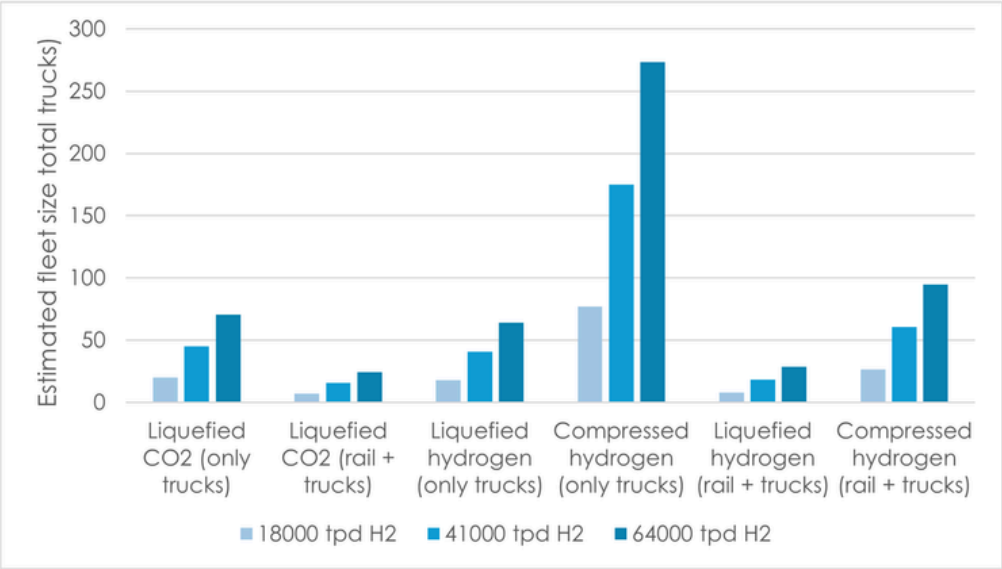


Figure 4: Comparison of the fleet size for liquid CO2, liquid H2 and compressed H2 transport in different $H_2$ throughput scenarios.

Fleet requirements are driven entirely by payload density:

- Compressed $H_2$ (0,8 t payload): high-demand scenario would require over 270 trucks, creating unfeasible traffic intensity (one truck every few minutes) on the 780 km route.
- Liquid $H_2$ (4 t payload): requires roughly twice the fleet of $CO_2$, requiring significant specialized cryogenic maintenance.
- Liquefied $CO_2$ (22 t payload): produces the lowest traffic density.

At an $H_2$-equivalent throughput of 41.000 t/year, $CO_2$ trucking needs over 50 trucks, but operating this fleet over 1.560 km round trips introduces significant risks.
So above this scale, a dedicated $CO_2$ pipeline becomes necessary, not only for potential long-term cost savings but primarily for supply-chain reliability.

## RECOMMENDATION

It is recommended to co-locate electrolysis with renewable power to eliminate $H_2$ transport entirely and in the near term, truck liquefied $CO_2$ to the eFuels synthesis plant (preferred over rail to avoid transloading complexity).

Pipeline infrastructure becomes compelling at higher throughput (e.g., over 450 t/day $CO_2$ over 500 km), driven by both cost and logistics-reliability considerations.

## ACKNOWLEDGMENTS

This fact sheet is designed for technical decision-makers (engineers, project developers, investors) while remaining accessible to policymakers. All figures are derived from conservative assumptions in the underlying technical assessment dated January 2026.

## REFERENCES

[1] E. Connelly, M. Penev, A. Elgowainy, and C. Hunter, 'Current Status of Hydrogen Liquefaction Costs'. US Department of Energy, Aug. 06, 2019.
[2] S. Jackson and E. Brodal, 'A comparison of the energy consumption for $CO_2$ compression process alternatives', IOP Conf. Ser. Earth Environ. Sci., vol. 167, p. 012031, Jul. 2018, doi: 10.1088/1755-1315/167/1/012031.
[3] S. Zs. Al Ghafri et al., 'Hydrogen liquefaction: a review of the fundamental physics, engineering practice and future opportunities', Energy Environ. Sci., vol. 15, no. 7, pp. 2690–2731, 2022, doi: 10.1039/D2EE00099G.

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https://github.com/IER-Hy4Daures/Fact-Sheets